SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I S.Praveen (192524239) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Pra

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1) Backtracking search - school Timetable Problem:

Given

Subjects:

* Math (M)
* Science (S)
* English (E)

Periods:

* Period 1 (P1)
* Period 2 (P2)
* Period 3 (P3)

Constraints:

i) Math cannot be in period 3.
ii) Science must be before English.
iii) Only one subject per second.
iv) English cannot be period 1.
v) Every subject must be assigned exactly one period.

Step 1: Assign Math

Try Math → Period 1

Constraint check:

- Math found in period 3
- Period 1 is free.

current Assignment:

- P1 = Math

Step 2: Assign science

Try science → Period 2

Constraint check:

- Period 2 is free
- Science before English

Current Assignment:

- P1 = Math
- P2 = Science

Step 3: Assign English

Try English → Period 3

Constraint check:

- English not in period 1
- Science before english (P2 < P3)
- Period 3 is free.

Current Assignment:

- P1 = Math
- P2 = Science
- P3 = English

} All subjects assigned

Constraint check Table:

Step	Assignment	Valid?	Reason
1	Math → P1	✓	Math not in P3
2	Science → P2	✓	Period free
3	English → P3	✓	English not in P1

Backtracking search tree:

```
Start
|
|--Math → P1
      |
      |-- Science → P2
               |
               |--English → P3 ✓ Solution
```

Final timetable:

Period 1	Math
Period 2	Science
Period 3	English

Uniform Cost Search (UCS) – Firefighting robot:

Grid (6x6)

```
S . . X . .
. X . X . .
. . R . X .
X . X . . .
. . . R X .
. X . . . G
```

Legend:

S = Start
G = Goal
X = Blocked
R = Risky cell (cost = 3)
. = Normal cell (cost = 1)

Coordinates:

- Start = (0,0)
- Goal = (5,5)

Step 1

Priority Queue

[(0, (0,0))]

Expand: (0,0)

Possible moves

- (0,1) cost = 1
- (1,0) cost = 1

Queue

[(1, (0,1)), (1, (1,0))]

Step 2:

Expand: (0,1)

New Node: (0,2) cost = 2

Queue: [(1,(1,0)), (2,(0,2))]

Step 3:

Expand: (1,0)

New Node: (2,0) cost = 2

Queue:
[(2,(0,2)), (2,(2,0))]

Step 4

Expand: (0,2)

New Node (1,2) cost = 3

Queue: ~~(3,~~ [(2,(2,0)), (3,(1,2))]

Step 5:

Expand: (2,0)

New Node: (2,1) cost = 3

Queue: [(3,(1,2)), (3,(2,1))]

Step 6:

Expand: (1,2)

New Node: (2,2) Risky cell
cost = 6

Queue: [(3,(2,1)), (6,(2,2))]

Step 7

Expand: (2,1)

New Node: (3,1) cost = 4

Queue [(4,(3,1)), (6,(2,2))]

Step 8:

Expand: (3,1)

New Node: (4,1) cost = 5

Queue: [(5,(4,1)), (6,(2,2))]

Step 9:

Expand: (4,1)

New Node: (4,2) cost = 6

Queue: [(6,(2,2)), (6,(4,2))]

Step 10:-

Expand:- (4,2)

New Node:- cost = 7

Queue:- [(6,(2,2)), (7,(5,2))]

Step 11:

Expand :- (5,2)

New Node:- (5,3) cost = 8

Queue :- [(6,(2,2)), (8,(5,3))]

Step 12:-

Expand:- (5,3)

New Node:- (5,4) cost = 9

Queue:- [(6(2,2)) (9,(5,4))]

Step 13:-

Expand :- (5,4)

New Node:- Goal (5,5)
cost = 10

Queue:- [(6,(2,2)), (10,(5,5))]

Step 14:-

Expand Goal

Goal Reached

Total cost calculation:

Move	Cost
10 Normal moves	10
Risk cells used	0

Total cost = 10

Optimal path:

S
↓
(1,0)
↓
(2,0)
→
(2,1)
↓
(3,1)
↓
(4,1)
→
(4,2)
↓
(5,2)
→
(5,3)
→
(5,4)
→
G